

Practical Demo: Constructing a Benchmark-Aware Bond Portfolio

From an investable universe to weights, trades, and stress tests

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This reading copy shows the calculations' results and figures. The HTML demo contains the full code in folded blocks and interactive scenario controls.

The desk question

How should a manager invest \$50 million near a bond benchmark while keeping active risk, concentration, and trading costs under control? Compare a simple cell sample with a constrained portfolio, then turn the chosen weights into a trade list and a decision memo.

Purpose and assumptions

An investment-grade manager has **\$50 million** to invest. The mandate is to earn more carry than a broad bond benchmark without taking an unintended duration, curve, sector, issuer, or liquidity bet. This demo turns that mandate into an auditable portfolio and an implementable trade list.

The security descriptions and market analytics are an **illustrative teaching universe**, not dated observations or executable quotes. In practice, the same table would come from a portfolio system, evaluated pricing service, dealer runs, and reference-data checks.

Workflow stage	In this case
Supplied	Bond terms, yields, spreads, durations, liquidity costs, benchmark and current weights
Constructed	A transparent cell-sampled portfolio and a constrained optimized portfolio
Diagnosed	Factor mismatches, tracking error, concentration, turnover, and transaction cost
Used for	Trade sizing, scenario P&L, and a portfolio recommendation

1 Define the Mandate

The optimized portfolio must be fully invested, hold no more than 18% in one security, keep at least 25% in Treasuries, cap corporate exposure at 45%, and stay within 0.05 years of

benchmark duration. These are teaching constraints; an actual mandate would also specify ratings, currencies, derivatives, cash, tax, ESG, and counterparty rules.

2 Assemble the Investable Universe

Yield and spread are annual percentages expressed as decimals. Duration and spread duration are in years; transaction cost is a one-way estimate in basis points of market value. KRD 2, KRD 5, and KRD 10 are simplified key-rate duration contributions per dollar invested.

	Sector	Cell	Yield
Bond			
UST 2Y	Treasury	Gov 1-4	0.0415
UST 5Y	Treasury	Gov 5-8	0.0435
UST 10Y	Treasury	Gov 9+	0.0455
Agency 3Y	Agency	Gov 1-4	0.0442
Agency 7Y	Agency	Gov 5-8	0.047
Bank A 3Y	Corporate	Corp 1-4	0.0505
Bank B 5Y	Corporate	Corp 5-8	0.0528
Utility A 7Y	Corporate	Corp 5-8	0.0555
Industrial A 10Y	Corporate	Corp 9+	0.0585
Industrial B 5Y	Corporate	Corp 5-8	0.054
Telecom A 10Y	Corporate	Corp 9+	0.06
Consumer A 7Y	Corporate	Corp 5-8	0.0562

	Spread (bp)	Duration	Spread duration
Bond			
UST 2Y	0	1.88	0
UST 5Y	0	4.45	0
UST 10Y	0	7.85	0
Agency 3Y	38	2.72	2.6
Agency 7Y	55	5.82	5.65
Bank A 3Y	92	2.65	2.55
Bank B 5Y	112	4.35	4.2
Utility A 7Y	138	5.65	5.45
Industrial A 10Y	165	7.35	7.05
Industrial B 5Y	125	4.18	4.02
Telecom A 10Y	182	7.1	6.75
Consumer A 7Y	148	5.45	5.25

	Trading cost (bp)	Benchmark weight
Bond		
UST 2Y	1	0.1
UST 5Y	2	0.13
UST 10Y	3	0.1
Agency 3Y	3	0.09
Agency 7Y	5	0.1
Bank A 3Y	6	0.07
Bank B 5Y	7	0.09
Utility A 7Y	9	0.08
Industrial A 10Y	12	0.07
Industrial B 5Y	8	0.07
Telecom A 10Y	14	0.05
Consumer A 7Y	10	0.05

3 Translate Holdings into Portfolio Exposures

i Read one row before constructing weights

Putting 10% of \$50 million in the UST 5Y means \$5 million of market value. Its 4.45-year duration contributes $\$5,000,000 \times 4.45 \times 0.0001 = \text{\$2,225 per bp}$. Its contribution to portfolio duration is $0.10 \times 4.45 = 0.445$ years. Dollar exposure adds across positions; security duration must first be weighted. The weighted average of individual yields is a carry proxy, not the portfolio's cash-flow internal rate of return.

For weights w_i and security exposure x_{ik} , portfolio exposure to factor k is $x_{P,k} = \sum_i w_i x_{ik}$. Dollar value of one basis point is $DV01 = V_P D_P(0.0001)$.

Yield (%)	4.944
Duration	4.842
Convexity	31.455
Spread duration	3.166
Treasury weight (%)	33.000
Corporate weight (%)	48.000
Largest position (%)	13.000
KRD 2	0.722
KRD 5	2.272
KRD 10	1.847
DV01 (\$/bp)	24209.500

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4 Build a Transparent Cell-Sampled Portfolio

Cell sampling first matches broad sector/maturity buckets, then chooses a representative liquid bond in each cell. It is easy to explain, but matching cell weights does not guarantee matching duration or key-rate exposure.

	Benchmark	Cell sampled
Bond		
UST 2Y	0.1	0.19
UST 5Y	0.13	0.23
UST 10Y	0.1	0.1
Agency 3Y	0.09	0
Agency 7Y	0.1	0
Bank A 3Y	0.07	0.07
Bank B 5Y	0.09	0.29
Utility A 7Y	0.08	0
Industrial A 10Y	0.07	0.12
Industrial B 5Y	0.07	0
Telecom A 10Y	0.05	0
Consumer A 7Y	0.05	0

5 Construct a Constrained Active Portfolio

Let $a = w - w_B$ be active weights and B the factor-exposure matrix. A compact factor-risk approximation is

$$TE^2 \approx (B'a)' \Sigma_F (B'a) + \sum_i a_i^2 \sigma_{\epsilon,i}^2.$$

The objective balances predicted tracking error, turnover, transaction cost, and a modest preference for carry:

$$\min_w TE^2 + \lambda_T \sum_i (w_i - w_i^{current})^2 + \lambda_C \sum_i c_i |w_i - w_i^{current}| - \lambda_Y w' y.$$

This is a teaching risk model. Production construction would use a validated factor covariance matrix, specific-risk forecasts, executable costs, lot sizes, and compliance-tested constraints. Here, factor and specific volatilities are annualized decimal return/rate units, so predicted tracking error is annualized. The objective coefficients are illustrative tuning weights, not estimated investor preferences or a guarantee of outperformance.

6 Compare the Alternatives

Check constraints before comparing yield. The cell sample is deliberately transparent, not constrained: putting a whole cell into one representative can breach the 18% security limit. The optimizer searches among feasible portfolios; it does not turn a higher assumed yield into a free excess return.

	Benchmark	Current	Cell sampled
Yield (%)	4.944	4.962	4.831
Duration	4.842	4.61	4.495
Convexity	31.455	28.776	27.932
Spread duration	3.166	3.187	2.242
Treasury weight (%)	33	30	52
Corporate weight (%)	48	52	48
Largest position (%)	13	12	29
KRD 2	0.722	0.802	0.705
KRD 5	2.272	2.235	2.17
KRD 10	1.847	1.572	1.62
DV01 (\$/bp)	24,209.5	23,048	22,473.5
Ex ante tracking error (%)	0	0.247	0.694
Turnover (%)	11	0	48
Estimated trading cost (\$)	7,250	0	30,750

	Optimized
Yield (%)	4.915
Duration	4.792
Convexity	31.024
Spread duration	3.068
Treasury weight (%)	34.805
Corporate weight (%)	45
Largest position (%)	12.416
KRD 2	0.732
KRD 5	2.278
KRD 10	1.781
DV01 (\$/bp)	23,959.5
Ex ante tracking error (%)	0.104
Turnover (%)	7
Estimated trading cost (\$)	3,462.414

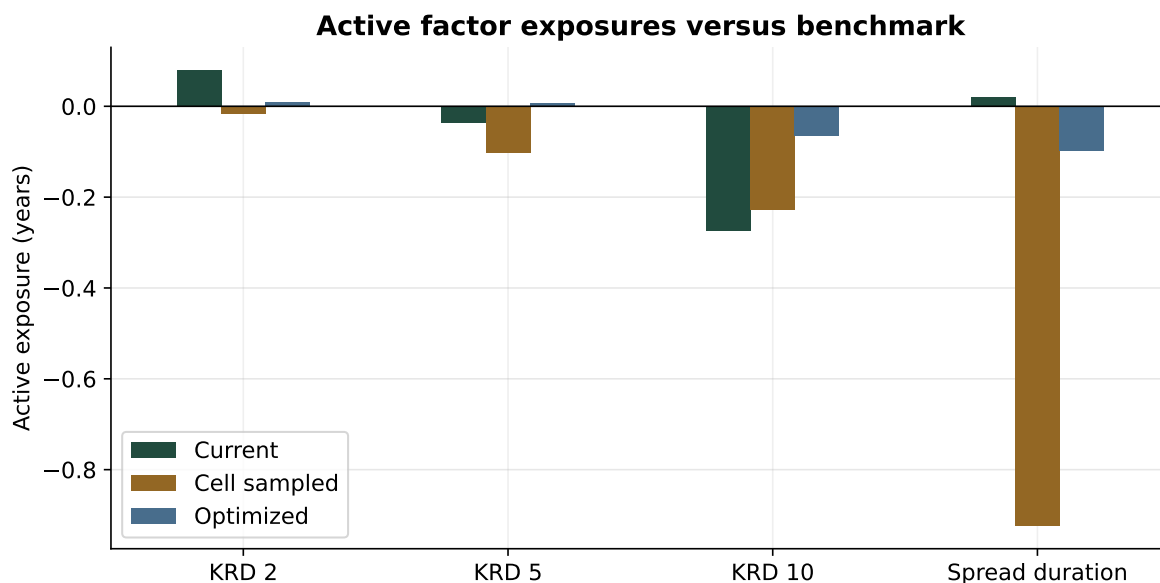


Figure 1: Optimization controls factor mismatches that cell weights alone can conceal

7 Turn Weights into Trades

Trades below are dollar market values. A real order list would convert market value to face amount using clean price, accrued interest, minimum denominations, round lots, and available dealer inventory.

	Sector	Cell	Current weight (%)
Bond			
UST 2Y	Treasury	Gov 1-4	12
UST 5Y	Treasury	Gov 5-8	10
UST 10Y	Treasury	Gov 9+	8
Agency 3Y	Agency	Gov 1-4	10
Agency 7Y	Agency	Gov 5-8	8
Bank A 3Y	Corporate	Corp 1-4	8
Bank B 5Y	Corporate	Corp 5-8	10
Industrial B 5Y	Corporate	Corp 5-8	8

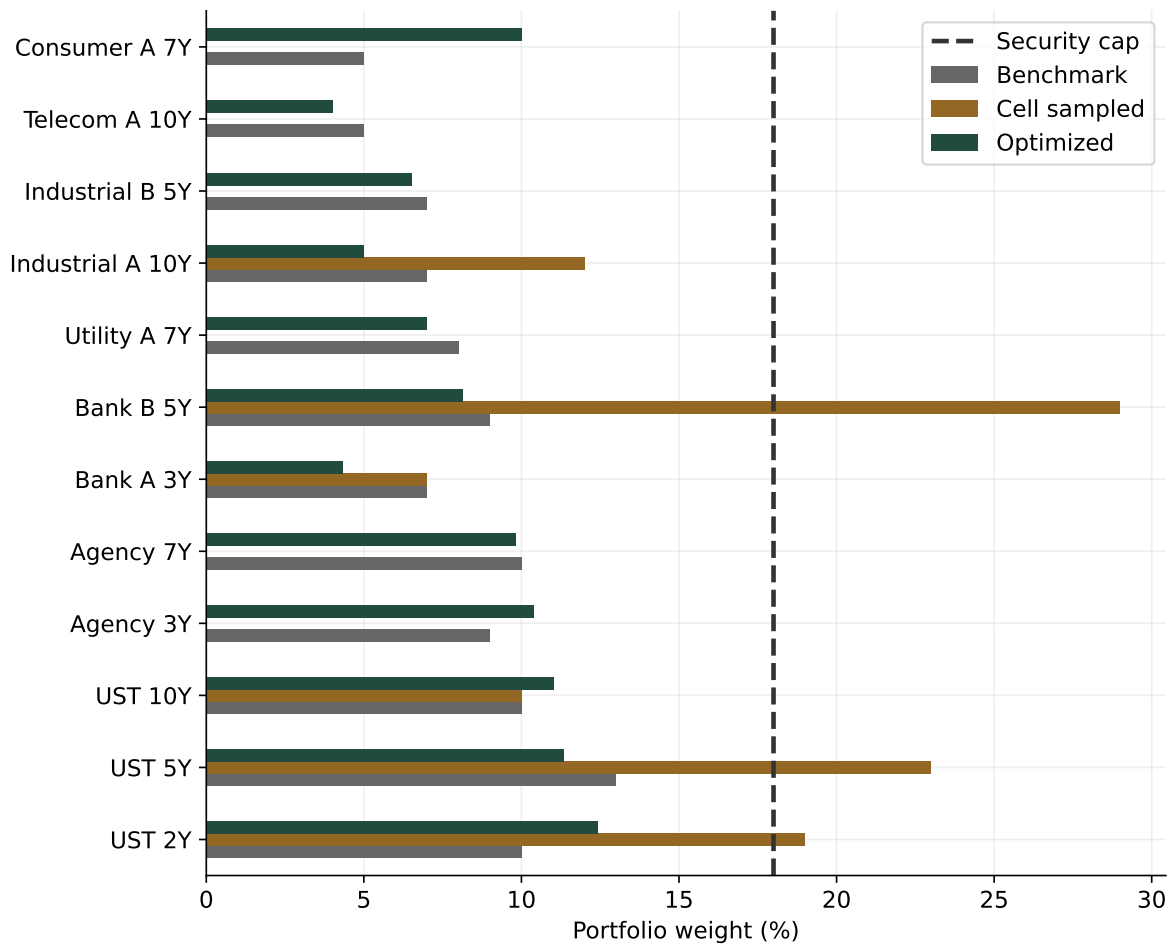


Figure 2: Security weights reveal concentration that an aggregate duration can hide.

Bond	Target weight (%)	Trade (\$mm)	Estimated cost (\$)
UST 2Y	12.416	0.208	20.778
UST 5Y	11.36	0.68	136.018
UST 10Y	11.029	1.515	454.369
Agency 3Y	10.38	0.19	57.047
Agency 7Y	9.815	0.907	453.707
Bank A 3Y	4.341	-1.83	1,097.806
Bank B 5Y	8.128	-0.936	655.032
Industrial B 5Y	6.531	-0.735	587.629

8 Stress-Test the Decision

For Treasury yield shift Δy_i and spread change Δs_i ,

$$\frac{\Delta P_i}{P_i} \approx -D_i \Delta y_i - SD_i \Delta s_i + \frac{1}{2} C_i (\Delta y_i)^2.$$

Scenario P&L (\$mm)	Benchmark	Current	Cell sampled
Parallel rates +50 bp	-1.191	-1.134	-1.106
Bear steepener	-1.515	-1.411	-1.375
Bull flattener	1.657	1.535	1.508
Credit selloff	-1.284	-1.326	-1.121

Scenario P&L (\$mm)	Optimized
Parallel rates +50 bp	-1.179
Bear steepener	-1.482
Bull flattener	1.624
Credit selloff	-1.225

8.1 Explore rate and spread shocks (HTML)

The nonparallel scenarios use each bond's three key-rate exposures, rather than assigning one maturity bucket to the entire bond. Parallel scenarios include convexity; twists use first-

order key-rate risk because a full cross-key-rate convexity matrix is not supplied. Agency spreads move by one quarter of the corporate spread shock in both the scenario table and the explorer.

The table above is the static PDF fallback. In HTML, change the parallel Treasury-rate and corporate-spread shocks to stress the optimized portfolio.

9 Validation and Recommendation

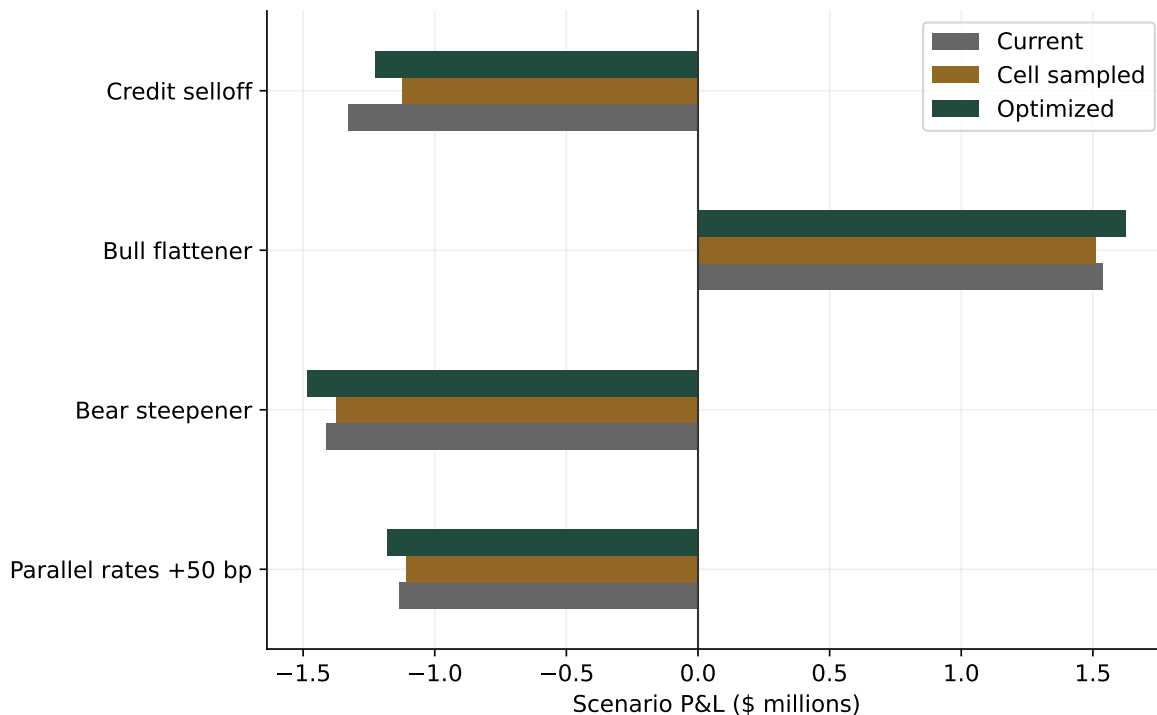


Figure 3: Compare stress losses before approving the trade list; these are scenarios, not return forecasts.

Optimization converged: True

Weights sum to	1.0000
Duration difference vs benchmark	-0.0500
Treasury minimum slack	0.0980
Corporate cap slack	0.0000
Position-cap slack	0.0558
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The optimizer is a disciplined starting point, not an automatic trade instruction. The manager should prefer it to the cell sample only if its incremental carry and factor control survive executable-price checks, lot rounding, issuer review, and scenario analysis. Rebalance when the benefit of restoring the mandate exceeds transaction costs and operational complexity.

10 Financial interpretation and market handoff

A portfolio can match aggregate duration and still carry different key-rate, spread, and issuer exposures. Check the whole constraint set and stress outcomes before comparing the carry proxy. The mini project below isolates the effect of a tighter corporate-sector limit while keeping the risk model fixed.

For a market-based extension, inspect bond terms and trade histories through [FINRA Fixed Income Data](#). Record identifiers, observation dates, price type and data permissions. Do not replace estimated risk exposures with unverified numbers. The embedded universe is the full fallback if an institutionally licensed inventory is unavailable.

Replace the embedded universe with a permitted bond inventory or evaluated-price snapshot. Add clean price and accrued interest, convert target market values to face amounts, impose issuer-level limits, and compare predicted tracking error with realized active returns over the next month.

11 Interactive explorer

Keep Treasury yields unchanged and move corporate spreads; agencies move by one quarter as much. Compare the three candidate portfolios before recommending a trade. These are instantaneous mark-to-market scenarios, excluding carry and transaction costs.

The HTML control selects a precomputed scenario. In the PDF, use the preceding figures and tables and rerun the displayed notebook inputs to explore other values.

12 Mini Project

Project brief. The investment committee reduces the corporate-sector cap from 45% to 35%. Starting from the same current holdings, solve for a new target using the same universe, benchmark, covariance matrix, transaction costs, and objective. Keep the 18% position cap, 25% Treasury minimum, and 0.05-year duration tolerance unchanged.

Required deliverable: a one-page decision memo with supporting tables.

1. Compare the 45% and 35% targets: corporate and Treasury weights, duration, carry proxy, annual predicted tracking error, turnover, and trading cost.
2. Show the 35% target and signed dollar trades for every bond. Reconcile buys and sells and confirm the new target satisfies every mandate constraint.
3. Compare all four supplied stress scenarios before and after estimated implementation cost. Do not add annual carry to an instantaneous stress.
4. Recommend a target under the revised mandate. Explain the credit-risk tradeoff and identify implementation checks still needed before trading.

Worked solution: revised mandate and trade memo

12.1 Solve the revised mandate

Only the corporate constraint changes. Both optimizations start from the same current holdings, so the cost comparison is between two alternative trade plans, not the cost of trading from one optimized target to the other.

	Corporate weight (%)	Treasury weight (%)	Duration
45% cap	45	34.8049	4.7919
35% cap	35	39.1888	4.7919

	Yield (%)	Annual predicted TE (%)
45% cap	4.9154	0.1037
35% cap	4.8261	0.1947

	One-way turnover (%)	Trading cost (\$)
45% cap	7	3,462.4136
35% cap	17	8,914.9832

12.2 Translate the target into trades

A positive trade buys market value; a negative trade sells it. These are dollar exposures, not face amounts or rounded trade tickets. Estimated costs are reported separately from the fully invested, gross target weights.

	Target weight (%)	Trade (\$mm)	Cost (\$)
Bond			
UST 2Y	14.2561	1.1281	112.8072
UST 5Y	12.903	1.4515	290.3007
UST 10Y	12.0297	2.0148	604.4504
Agency 3Y	12.899	1.4495	434.8478
Agency 7Y	12.9122	2.4561	1,228.0484
Bank A 3Y	1.9155	-3.0422	1,825.3453
Bank B 5Y	6.0363	-1.9818	1,387.2944
Utility A 7Y	5.2961	-0.8519	766.7346
Industrial A 10Y	5	-0	0.0029
Industrial B 5Y	4.4117	-1.7941	1,435.3103
Telecom A 10Y	4	-0	0.0002
Consumer A 7Y	8.3403	-0.8298	829.841

12.3 Compare stress outcomes and decide

The net exhibit deducts each alternative's one-time implementation cost from its instantaneous mark-to-market stress P&L. It assumes trading at the initial prices followed by the shock. It is not a one-year return forecast.

Gross stress P&L (\$ millions)

	45% cap	35% cap
Scenario		
Parallel rates +50 bp	-1.1786	-1.1784
Bear steepener	-1.4821	-1.4586
Bull flattener	1.6238	1.6081
Credit selloff	-1.2249	-1.0475

Stress P&L after implementation cost (\$ millions)

	45% cap	35% cap
Scenario		
Parallel rates +50 bp	-1.182	-1.1873
Bear steepener	-1.4856	-1.4675
Bull flattener	1.6204	1.5992
Credit selloff	-1.2283	-1.0564

The tighter target changes gross credit-selloff P&L by **\$177,386.68** and implementation cost by **\$5,452.57** relative to the 45% plan. These compare a conditional stress benefit with a one-time cost; they do not establish expected profit.

Recommendation. Use the 35% target as the starting proposal because the 45% target is no longer permissible under the revised mandate. Lower corporate exposure reduces the specified credit-selloff loss, but may reduce the carry proxy, increase benchmark-relative risk, and require additional turnover. Matching duration alone does not remove those tradeoffs.

Before execution, verify prices, accrued interest, issuer aggregation, liquidity, and lot sizes; convert market-value trades to face amounts and rerun constraints on rounded tickets. If transaction costs must be funded inside the \$50 million, reserve that cash and reoptimize: gross weights that sum to one do not also pay fees. The covariance matrix, spreads, and expected trading costs are teaching inputs, and the stress scenarios carry no assigned probabilities.